I. Correctness and Parameter Analysis

A. Correctness

Let $\alpha = \frac{e^{1/5}}{\sqrt{2\pi}}$. Probabilities in this correctness analysis include the sampling of (s, e) during key generation as well as preprocessing. We use the standard subgaussian norm bound for the ideal discrete-Gaussian samples and apply it directly to independent sums; no convolution is replaced by an exact spherical discrete Gaussian. Concretely, for independent centered samples x_k with subgaussian parameters σ_k, the coefficientwise moment-generating-function bound and independence give subgaussian parameter $(\sum_k \sigma_k^2)^{1/2}$ for $\sum_k x_k$; the standard vector-norm tail bound with $\alpha = \frac{e^{1/5}}{\sqrt{2\pi}}$. and $N \geq 16\lambda$ is at most $2^{-\lambda}$. In particular, for $a = |\mathsf{act}| \leq a_{\max}$, except with probability at most $6 \cdot 2^{-\lambda}$, all six bounds

$$\left\|\sum_j y_j\right\|_2 \leq \alpha\sqrt{aN}\,\sigma_2, \qquad \left\|\sum_j e'_j\right\|_2 \leq \alpha\sqrt{aN}\,\sigma_2,$$
$$\left\|\sum_j r_{j,1}\right\|_2, \left\|\sum_j r_{j,2}\right\|_2 \leq \alpha\sqrt{aN}\,\sigma_r,$$
$$\|s\|_2, \|e\|_2 \leq \alpha\sqrt{N}\,\sigma_1$$

hold simultaneously when $N \geq 16\lambda$. The indices in these sums range over act. Since $\|c\|_1 = \tau$, Young's inequality for negacyclic convolution gives

$$\|sc\|_2 \leq \tau\|s\|_2, \qquad \|ec\|_2 \leq \tau\|e\|_2.$$

This avoids treating sc or ec as an independent spherical Gaussian.

To support every active set of size at most $a_{\max}$, define the conservative bounds

$$\beta_z(a_{\max}) = \alpha\sqrt{N}\left(\sqrt{a_{\max}}\,\sigma_2 + \tau\sigma_1\right),$$
$$\beta_r(a_{\max}) = \alpha\sqrt{a_{\max}N}\,\sigma_r,$$
$$\beta_{\widetilde{e}}(a_{\max}) = \alpha\sqrt{N}\left(\sqrt{a_{\max}}(\sigma_2 + \sigma_r) + \tau\sigma_1\right) + \frac{q\tau}{2q_1}\sqrt{N},$$
$$\beta_\Delta(a_{\max}) = \frac{q_2}{q}\beta_{\widetilde{e}}(a_{\max}) + \sqrt{N}.$$

The verifier uses $\beta_z = \beta_z(a_{\max})$, $\beta_r = \beta_r(a_{\max})$, and $\beta_\Delta = \beta_\Delta(a_{\max})$.

For an honest execution, pairwise-mask cancellation and Shamir reconstruction give

$$z = \sum_{j\in\mathsf{act}} (y_j + \lambda_{\mathsf{act},j}s_jc + m_j)$$
$$= \sum_{j\in\mathsf{act}} y_j + sc, \qquad r = \sum_{j\in\mathsf{act}} r_{j,2}.$$

Let

$$\eta_1 = \widehat{b} - \mathsf{Lift}_1(b), \qquad \|\eta_1\|_\infty \leq \frac{q}{2q_1},$$

and define the residual, using canonical centered representatives in $\mathcal{M}_q$, by

$$\widetilde{e} = \sum_{j\in\mathsf{act}} (e'_j + r_{j,1}) + ec - \eta_1 c.$$

For $X = az + h_{\mathsf{sid}}r - \mathsf{Lift}_1(bc_{\mathbb{Z}})$, where the product is formed in R_{q_1} before lifting, the honest aggregate token satisfies

$$C_{\mathsf{act}} = \lfloor X + \widetilde{e}\rceil_{p_2}.$$

The bounds above and $\|\eta_1 c\|_2 \leq (q\tau/(2q_1))\sqrt{N}$ imply $\|\widetilde{e}\|_2 \leq \beta_{\widetilde{e}}(a_{\max})$. The modulus-switching difference bound then gives

$$\|\Delta\|_2 \leq \frac{q_2}{q}\|\widetilde{e}\|_2 + \sqrt{N} \leq \beta_\Delta(a_{\max}).$$

Finally, the definition of Δ yields

$$C' = \Delta + \lfloor X\rceil_{p_2} = C_{\mathsf{act}},$$

so the verifier recomputes the honest challenge. In this construction, at least one honest signer belongs to every admissible active set because $|\mathcal{C}| < t \leq |\mathsf{act}|$. A successful honest signing execution has therefore bound the exact context before its response collection completes. The persistent binding read matches the context reconstructed above. Hence the added check introduces no further cryptographic correctness error with the ideal available service. An implementation's probability of an unavailable or invalid read must be accounted for separately; correctness does not promise progress after adversarial burns that prevent a signing execution from completing.

A union bound over the six displayed subgaussian events gives

$$\varepsilon^{\mathsf{pt}}_{\mathsf{corr}}(\lambda) \leq 6 \cdot 2^{-\lambda} + \varepsilon_{\mathsf{samp}}(\lambda),$$

where $\varepsilon_{\mathsf{samp}}$ is the joint statistical distance of an implemented Gaussian sampler from the ideal samples and is zero in the ideal distributional model. In the pointwise bound, this joint distribution contains (s, e) and all samples used by the fixed active set. In the uniform bound, it contains (s, e) and the complete preprocessing family for every signer from which the active set may be selected. If only per-sample distances are available, $\varepsilon_{\mathsf{samp}}$ includes their composition over the corresponding batch. A concrete implementation is required to make $\varepsilon_{\mathsf{samp}}(\lambda)$ negligible. This is a pointwise bound for each fixed admissible act chosen independently of the preprocessing randomness. For active-set selection after observing the public tokens, let

$$S(n, t, a_{\max}) = \sum_{a=t}^{a_{\max}} \binom{n}{a}.$$

For each admissible act, let $(\sigma_{\mathsf{act}}, \mathcal{B}_{\mathsf{act}})$ be the result of an honest counterfactual execution from a separate copy of the same fresh post-preprocessing state, using the common sampled key and token family. Each copy binds only its own context. The event below means that every such choice would succeed if selected; only one choice is executed in the real protocol. A union bound on the Gaussian norm events before choosing the active set gives the uniform event

$$\Pr\left[\begin{array}{c}\forall \mathsf{act} \subseteq [n], \quad t \leq |\mathsf{act}| \leq a_{\max} : \\ \mathsf{Vf}^{\mathcal{B}_{\mathsf{act}}}(\mathsf{vk}, \mathsf{msg}, \sigma_{\mathsf{act}}) = 1\end{array}\right]$$
$$\geq 1 - 6S(n, t, a_{\max})2^{-\lambda} - \varepsilon_{\mathsf{samp}}(\lambda).$$

Thus adaptive post-preprocessing selection uses $\varepsilon^{\mathsf{unif}}_{\mathsf{corr}} \leq 6S(n, t, a_{\max})2^{-\lambda} + \varepsilon_{\mathsf{samp}}$. The supported parameter regime

must restrict $(n, t, a_{\max}, N)$ so that the chosen pointwise or uniform error target is negligible.

B. Parameter Analysis

Inputs And Conventions. We use $R_q = \mathbb{Z}_q[X]/(X^N + 1)$, centered representatives, and $D_\sigma(x) \propto \exp(-\pi x^2/\sigma^2)$. Every σ in this profile, including σ_{td}, denotes a Gaussian width. The corresponding standard-deviation parameter is $\sigma/\sqrt{2\pi}$. Fix $n = t = a_{\max} = 1024$, a challenge-query budget $Q = 2^{60}$, and $\epsilon = 2^{-320}$ for checking the smoothing and trapdoor inequalities. The statistical parameter does not itself certify the error of an implemented sampler. The revised prime modulus satisfies $q \equiv 1 \pmod{2N}$.

Parameter	TT-128-R1
N	4096
q	18014398509309953
τ	32
$(\sigma_1, \sigma_2, \sigma_r)$	$(2^6, 2^{40}, 2^{31})$
σ_{td}	4349014
(p_1, p_2)	$(38, 42)$
(q_1, q_2)	$(65536, 4096)$

Here $q_i = \lfloor q/2^{p_i} \rceil$ exactly. Although p_1, p_2 increase, the larger modulus gives larger compressed alphabets and thus less relative compression than the retired profiles.

Exact Verification Thresholds. Use $\alpha = e^{1/5}/\sqrt{2\pi}$ and the preceding correctness formulas to obtain ideal bounds β_z^{ideal}, β_r^{ideal}, and $\beta_\Delta^{\text{ideal}}$. Round their squares outwards:

$$K_z = \lceil (\beta_z^{\text{ideal}})^2 \rceil, \quad K_r = \lceil (\beta_r^{\text{ideal}})^2 \rceil, \quad K_\Delta = \lceil (\beta_\Delta^{\text{ideal}})^2 \rceil.$$

The exact constants are

$$\begin{aligned} K_z &= 120391959247771758179861769 4083, \\ K_r &= 45925887769224658775284 17, \\ K_\Delta &= 142861. \end{aligned}$$

The verifier compares the integer squared norms with these integers. For this profile, use the actual radii $\beta_z = \sqrt{K_z}$, $\beta_r = \sqrt{K_r}$, and $\beta_\Delta = \sqrt{K_\Delta}$ throughout extraction. Numerically,

$$\begin{aligned} \beta_z &\simeq 1.097232697506649 \cdot 10^{15}, \\ \beta_r &\simeq 2.143032612192933 \cdot 10^{12}, \\ \beta_\Delta &\simeq 377.9695754952. \end{aligned}$$

Substituting these actual radii into the complete extraction formula gives

$$\begin{aligned} \beta_\delta &\simeq 4.169080475712151 \cdot 10^{15}, \\ \beta_{\mathsf{SIS}} &\simeq 4.711361666570529 \cdot 10^{15} \\ &\simeq 0.2615331099806451q < (q-1)/2. \end{aligned}$$

This removes the previous universal public-column short-solution obstruction; it does not prove normalized ring-SIS hardness.

Strict Width Checks And Correctness. We use the strict Hint-RLWE relation in TalonG's Theorem 1, retaining τ^2 rather than replacing the relaxed asymptotic factor by one:

$$\sigma = \left(\frac{2}{\sigma_1^2} + \frac{2Q\tau^2}{\sigma_2^2} \right)^{-1/2} \simeq 20.2385770251.$$

A sufficient upper bound on $\sqrt{2}\eta_\epsilon(\mathbb{Z}^N)$ is $\sqrt{2\ln(2N(1+1/\epsilon))/\pi} \simeq 12.1220175590$. The selected ephemeral and preimage widths satisfy

$$\begin{aligned} \sigma_2 &> 2\sqrt{N} q^{1/2+1/N} \simeq 1.7337581266 \cdot 10^{10}, \\ \sigma_r &> 1.17\sqrt{q}\sqrt{\ln(4N(1+1/\epsilon))/\pi} \simeq 1.3480523948 \cdot 10^9. \end{aligned}$$

The second inequality is sufficient for the cited smoothing bound only when the generated trapdoor satisfies $\|\widetilde{B}\|_{\mathsf{GS}} \le 1.17\sqrt{q}$. In Falcon specification v1.2, the sampling choice $1.17\sqrt{q/(2N)}$ uses a standard-deviation parameter, whereas this profile uses Gaussian widths.[1] The corresponding width and our rounded candidate are

$$\begin{aligned} \sigma_{\mathsf{td}}^{\mathsf{Falcon}} &= \sqrt{2\pi}\, 1.17\sqrt{q/(2N)} \\ &\simeq 4349013.011830253, \\ \sigma_{\mathsf{td}} &= \lceil \sigma_{\mathsf{td}}^{\mathsf{Falcon}} \rceil = 4349014. \end{aligned}$$

This conversion replaces the former width 1735006; it is not a universal trapdoor-width lower bound or a proof that a particular TDGen is efficient. At the chosen (N, q), the generator must still establish its acceptance probability, Gram–Schmidt bound, and public-key distribution, and Pre must establish its conditional sampling guarantee. Correcting σ_{td} does not change $\sigma_1, \sigma_2, \sigma_r$ or the displayed correctness and extraction radii. In particular, σ_r is not rescaled mechanically. These scalar inequalities and the verification thresholds are numerical checks, not a certification of those trapdoor guarantees. Since $n = t = a_{\max}$, the active-set union factor is one. The six-event ideal correctness tail is at most

$$6\exp\left(N \left[\frac{1}{5} + \frac{1 - e^{2/5}}{2} \right] \right) \simeq 2^{-268.72391}$$

per signing execution, plus the joint sampler error for that execution. This is not an experiment-wide sampler-error bound.

Estimator Results and Their Scope We ran the unmodified Lattice Estimator at commits `5ba00f5` (the supplied TalonG reference) and `53da598` (a second version check), using SageMath 10.9, the classical MATZOV model, and the explicit `gsa` shape model. Gaussian widths are converted to standard deviations by division by $\sqrt{2\pi}$. The LWE and SIS inputs are unchanged by the trapdoor-width correction, so their recorded outputs below are retained. The NTRU run would use the corrected standard-deviation parameter $4349014/\sqrt{2\pi}$, together with an analysis of the screened key distribution defined below. We defer this numerical evaluation. The conditional theorem retains the screened-key indistinguishability assumption; deferring an attack-cost estimate does not discharge it. The generic coefficient-lattice mappings are LWE $(n, m) = (N, N)$, Euclidean SIS $(n, m) = (N, 4N)$, and circulant NTRU $(n, m) = (N, N)$. These mappings are heuristic for the structured ring instances used here, including normalized ring-SIS.

Problem	MATZOV bits	BKZ block
Hint-RLWE (BDD)	301.11	956
SIS (full bound)	299.04	949
NTRU (uSVP)	not evaluated	—

[1] P.-A. Fouque et al., *Falcon: Fast-Fourier Lattice-based Compact Signatures over NTRU*, specification v1.2, 2020; https://falcon-sign.info/falcon.pdf.

The first two rows are the lowest finite, usable outputs in the recorded MATZOV runs for those problems. With the classical ADPS16 model, LWE uSVP costs $2^{280.904}$ and SIS costs $2^{277.108}$. These are model-specific work estimates, not bounds on distinguishing advantage. The former NTRU estimate of 971.24 bits at block size 3335 used the superseded trapdoor-width input and was also an extrapolation beyond the fitted block-size range. It is withdrawn for this corrected profile. Other recorded NTRU calls encountered model limits or returned non-finite outputs. The recorded BKW call failed numerically. All diagnostics are retained in the reproducibility package; a failed call or infinite output does not certify security or establish the minimum over all attacks.

Interpretation of Heuristic Estimates. The cost models and generic coefficient-lattice mappings above are used as explicitly heuristic evidence about known attacks. They are not claimed as proven lower bounds against all algorithms. Such estimates can guide a candidate parameter choice without a tight concrete security reduction, provided that the instance distribution, attack coverage, cost model, and extrapolations are stated. In contrast, simulation identities, extraction bounds, and statistical-distance composition in the conditional theorem require mathematical justification. An unproved primitive property must be identified as an assumption rather than inferred from a favorable estimator output. In particular, the omitted NTRU estimate leaves this profile without a complete numerical assessment of all assumptions used by the proof.

Additional Screens And Communication. The challenge entropy is $\tau + \log_2 \binom{N}{\tau} \simeq 298.1615854$ bits. The correction radius is strictly below $(q_2/2)\sqrt{N}$. For only the fresh, uniform-C, $z = r = 0$ strategy with a fresh challenge-oracle input, the finite-alphabet Chernoff bound is less than 2^{-30272} per trial. This does not bound a fixed or adversarially selected C, or general forgeries. Simple fixed-width encoding uses 51, 42, and 10 bits per coefficient of z, r, Δ, respectively. Their total is 52,736 bytes (51.5 KiB), excluding the challenge and metadata. Encoding full a and compressed b takes 35 KiB; one uncompressed public token polynomial takes 27 KiB. A canonical sparse challenge representation takes 52 bytes. These are encoding budgets, not measured protocol bandwidth or runtime results.

Remaining Proof And Implementation Obligations. The state-assisted theorem eliminates E_1 in the ideal service model, but its E_0 term still incurs label-guessing and forking losses. No end-to-end 128-bit claim follows from the estimator outputs alone. Such a concrete claim requires explicit experiment-wide budgets, including L, T, PRF and sampler errors, supported primitive advantage bounds, and a realization of the trusted service. A trapdoor implementation must supply the required guarantees at this modulus and dimension. Changing the supported participant sets, query budget, or widths requires recomputation of both correctness and security conditions.

II. Security of the Abstract Framework

A. Proof of Lemma 1 (State Invariant)

Proof. Order the atomic transactions by their linearization points. Initially no response exists. Token transactions create states only while the identifier is fresh and return the same record on a repeated query. The first successful response records its context and consumes its signer's state. Each subsequent successful response checks equality with that recorded context and consumes a different live state. A mismatch or burn makes every remaining state unavailable, and the burned state is terminal. Every subsequent prescribed transaction preserves the binding $\mathsf{UseCtx}[\mathsf{sid}] = \mathsf{ctx}_0$, including transactions that burn the identifier or erase private token state. Induction on this transaction order proves both response assertions and preservation of the binding. Concurrent requests do not alter the argument because response release belongs to the same atomic transaction as consumption. □

B. Proof of Lemma 1A (Binding Excludes Incompatible-Context Acceptance)

Proof. An OT win requires a recorded $\mathsf{ctx}_0 \neq \perp$, at least one released honest response for it, and a different terminal $\mathsf{ctx}^\star$ under the same identifier. Lemma 1 and binding persistence imply that the terminal binding read returns $\mathsf{Bound}(\mathsf{ctx}_0)$, even after a burn. The final-state convention ensures the winning predicate and read refer to the same record. The state-assisted verifier accepts this read only if $\mathsf{ctx}^\star = \mathsf{ctx}_0$, a contradiction. This argument also covers an old active set changed to a different set and any change of the message or reconstructed aggregate. It does not use lattice hardness and makes no assertion about a stateless verifier. □

C. Proof of Lemma 2 (Aggregate-Equivalent Views)

Proof. Fix the same initial signer and identifier-service states, shared random-oracle functions, PRF values, call order, and external state-service events in both executions. Compare any two admissible mapping families $\mathbf{M}$ and $\mathbf{M}'$ from Section IV-C (Flexible Active-Set Selection) of the main text. In each call, both mappings pass the same structural and indexed-token checks and yield the same C_{act}. All subsequent context and consumption checks therefore have identical outcomes. Whenever a response is released, x_{ctx}, c, and

$$z_j = y_j + \lambda_{\mathsf{act},j} s_j c + m_j$$

depend on the mapping only through the common context. The second response component is the same stored $r_{j,2}$. Thus the response or rejection and the resulting state transition agree pointwise. Induction over the calls preserves the entire response vector and the state at every stopping prefix. For any fixed complete response collection, Combine uses its mapping only for structural validation and recomputing the common aggregate; its output or rejection is therefore unchanged as well. This proves exact joint distributional equality. The argument requires neither a common mapping accepted by all honest signers, equality of unrounded token sums, nor an assumption that different mappings are computationally difficult to find. □

D. Proof of Lemma 3 (Mask Distribution And Partial Responses)

Proof. For each unordered pair of honest active signers, the difference of its two independent directed random-function

outputs is uniform in R_q. The incidence map of the complete graph on J maps these edge differences onto the zero-sum submodule of $R_q^{|J|}$. Surjectivity follows already from a spanning tree: assign its leaf edges successively to realize any prescribed zero-sum vector. All fibers have the same size, since this is a homomorphism of finite additive groups. Contributions from corrupted-pair edges translate the image by a vector whose sum is $-m_K$. Thus, conditional on the corrupted keys and the context, $(m_j)_{j\in J}$ is uniform on

$$\left\{ (u_j)_{j\in J} \in R_q^{|J|} : \sum_{j\in J} u_j = -m_K \right\}.$$

Projection onto any proper set of coordinates is surjective with equally sized fibers. For $|J| = 1$, the mask is fixed and there is no nonempty proper subset. Domain separation, injective context encoding, and Lemma 1 permit this argument independently for each new context; repeated requests reuse the same record. □

E. Full Proof of Theorem 2 (Identifiers Without A Completed Context)

Proof. Work first with ideal Gaussian sampling and the randomized honest-pair masks of Lemma 3. Guess an index $I \leftarrow [L]$ in the order of first relevant identifier-label evaluations. Call its identifier $\mathsf{sid}^\dagger$. If this index is unused, the reduction fails. Require the terminal identifier to equal $\mathsf{sid}^\dagger$ and abort if a response would complete a context under this identifier. Before any distribution change, this restriction preserves exactly $1/L$ of the randomized-mask game's incomplete-context success probability: no execution counted by $\mathsf{Win}^{\mathsf{core}}_{\mathsf{PA}} \wedge \neg\mathsf{Complete}(\mathsf{sid}^\star)$ ever triggers this abort when the guess is correct. This selection uses identifiers, including previously exposed ones, without guessing a message or a future active set.

Every binding read, including adversarial public reads, is answered from the simulator's own current UseCtx table. The table contains only contexts already supplied in successful response queries; no secret or trapdoor is needed to answer a read. Reads neither authorize messages nor add labels to the L count. All hybrids below maintain these tables and their transitions. Thus the added interface is simulated together with the original oracle view, including adaptive decisions based on reads.

Generation gate and failure accounting. We first work with the ideal capped TDGen. At the first label evaluation of each nondistinguished identifier, run an auxiliary capped generation and abort if it returns $\perp$. In the initial gated hybrid, discard any successful output and sample the published label independently as $h \leftarrow R_q$. Tokens and responses are still generated by the original algorithms. The added abort probability and the comparison with implemented generation are charged once to $\epsilon_{\mathsf{badTD}}$.

Screened-key hybrid. Next, replace the independently uniform label by the public component of the successful auxiliary output (h, td). Record this label consistently in the label-oracle table. Its successful-output law is $\mathcal{D}_h^{\mathsf{acc}}$. Retain td internally, but do not use it in this hybrid. All token and response computations still follow the original algorithms.

For each label replacement, a screened-key distinguisher reproduces the generation gate and uses its challenge h as the published label. It generates the signing keys itself and answers all protocol queries without a trapdoor for the challenge label. The screened-key assumption therefore bounds the computational loss by ϵ_{key} per label, and by $(L-1)\epsilon_{\mathsf{key}}$ in total. The distinguished label remains uniform.

Token and opening hybrid. Keeping the generated labels and trapdoors fixed, replace each new nondistinguished honest token by an independent uniform $W_j \leftarrow R_q$. For comparison, the original forward sampler satisfies

$$W_j = ay_j + e'_j + f_h(r_{j,1}, r_{j,2}),$$

where $(y_j, e'_j) \leftarrow D^2_{\sigma_2}$ and $(r_{j,1}, r_{j,2}) \leftarrow D^2_{\sigma_r}$.

Replacing the image $f_h(r_{j,1}, r_{j,2})$ by uniform, while preserving the exact conditional preimage law, costs at most δ_{img} per token. Implementing that conditional sampler with Pre adds at most $\delta_{\mathsf{GPV}} + d\nu$. Thus the joint token–opening comparison costs at most ϵ_{pre} per token, including its later opening. The aggregate statistical loss is at most $T\epsilon_{\mathsf{pre}}$.

After the image replacement, W_j is uniform independently of (y_j, e'_j). These variables and the opening may therefore be deferred until a response is needed, using

$$(r_{j,1}, r_{j,2}) \leftarrow \mathsf{Pre}(h, \mathsf{td}, W_j - ay_j - e'_j).$$

With the ideal conditional sampler, this deferral is an exact distributional rewriting; its approximation error is already included in ϵ_{pre}. Repeated token queries return the same recorded token. Erased unused records need never be opened.

The computational label replacement, statistical token replacement, and generation comparison are together charged to ϵ_{TD}.

Crucially, the distinguished identifier is treated differently: sample all its tokens and $(y_j, e'_j, r_{j,1}, r_{j,2})$ honestly under its uniform h. For each permitted response return a uniform z_j and the genuinely stored $r_{j,2}$. Lemma 3 proves the joint distribution correct until completion, where this restricted game aborts. No trapdoor for the distinguished h and no equivocation of a target token is required. Terminal acceptance in this experiment uses $\mathsf{Vf}_{\mathsf{lat}}$; binding reads made by the adversary are still answered from the simulated service table. The same argument holds if its active set was chosen after seeing all its tokens.

Explicit reduction to batch Hint-RLWE. Let G_{real} denote the preceding simulation hybrid with $B = as + e$. Let G_{uniform} denote the same hybrid with an independent uniform B. Both games retain the same hidden (s, e) in all hints, and use the same token samplers, state transitions, and abort rules.

We construct a distinguisher $\mathcal{B}_{\mathsf{Hint}}$ whose input is

$$X = \left(a, B, \{(c_i, U_i, V_i)\}_{i=1}^{Q}\right),$$

drawn from either batch Hint-RLWE distribution defined above. The distinguisher does not receive s, e, or any honest Shamir share. It runs $\mathcal{A}$ and simulates its view as follows.

1) *Initialization and corrupted keys.* Run $\mathcal{A}_0$ to obtain the static corruption set $\mathcal{C}$ and its continuation state.

Return 0 if the corruption set is invalid. Sample a fresh coefficient-wise Shamir sharing of zero, retaining only the shares $\{s_k\}_{k\in\mathcal{C}}$. Because $|\mathcal{C}| < t$, their joint distribution is identical to the restriction of a sharing of any secret s.

Generate the PRF material and randomized honest–honest masks as in the preceding hybrid. Give $\mathcal{A}_1$ the corrupted secret-key material and

$$\mathsf{vk} = (a, \lfloor B \rceil_{p_1}, t, n, a_{\max}).$$

Follow the same distinguished-label selection convention and initialize all oracle and identifier-state tables.

2) *Labels and challenge queries.* Sample the distinguished label uniformly. For every nondistinguished label, run the capped TDGen and retain its successful pair (h, td). Return 0 on a generation abort. Repeated label queries receive the same recorded output.

   Before the terminal output, assign the ith input hint to the ith distinct challenge-oracle input, and answer that input with c_i. Repeated inputs reuse their assigned index. Index hints by oracle inputs, rather than by challenge values. Ordinary queries to the remaining random-oracle domain are handled by lazy sampling.

   If terminal verification makes a fresh challenge query, answer it with an independent uniform challenge. No hint is required for this terminal-only query.

3) *Token queries.* Apply the token-state checks and idempotence rules of the preceding hybrid. For a new honest token under the distinguished identifier, sample

$$(y_j, e'_j) \leftarrow D^2_{\sigma_2}, \quad (r_{j,1}, r_{j,2}) \leftarrow D^2_{\sigma_r},$$

and compute

$$W_j = ay_j + e'_j + f_h(r_{j,1}, r_{j,2}).$$

   Store the corresponding private record.

   For a new honest token under a nondistinguished identifier, sample $W_j \leftarrow R_q$ and store its unopened record. In both cases, return the public token (j, sid, W_j).

4) *Response queries.* Perform the structural, recorded-token, aggregate, challenge, and bound-context checks of the preceding hybrid. Maintain the same rejection and burn behavior.

   For the current context, use J, s_K, and m_K as defined in Lemma 3. The values s_K and m_K are computable from the corrupted key material.

   Under the distinguished identifier, return

$$\sigma_j = (z_j, r_{j,2}), \quad z_j \leftarrow R_q,$$

   using the genuinely stored $r_{j,2}$. Return 0 if a successful response would complete the distinguished context.

   For a nondistinguished nonfinal honest responder, sample $(y_j, e'_j) \leftarrow D^2_{\sigma_2}$, choose $z_j \leftarrow R_q$, and sample

$$(r_{j,1}, r_{j,2}) \leftarrow \mathsf{Pre}(h, \mathsf{td}, W_j - ay_j - e'_j).$$

   Store y_j and z_j, and return $(z_j, r_{j,2})$.

   For the last honest responder j, retrieve the hint assigned to the context's challenge input and compute

$$\begin{aligned} w_j^{\mathsf{res}} &= W_j - aU_i + Bc_i - V_i, \\ (r_{j,1}, r_{j,2}) &\leftarrow \mathsf{Pre}(h, \mathsf{td}, w_j^{\mathsf{res}}), \\ z_j &= U_i - s_K c_i - m_K + \sum_{k\in J\setminus\{j\}} (y_k - z_k). \end{aligned} \tag{1}$$

   Return $(z_j, r_{j,2})$. Each successful response performs the prescribed atomic binding, consumption, erasure, and $\mathsf{RespSet}$ update.

5) *Binding and burn queries.* Answer binding reads from the current UseCtx table. Process burn queries using the same erasure and persistence rules. These operations require no honest secret share.

6) *Terminal decision.* On $\mathcal{A}$'s terminal output, evaluate $\mathsf{Win}^{\mathsf{core}}_{\mathsf{PA}}$ using $\mathsf{Vf}_{\mathsf{lat}}$ and the recorded state. Output 1 exactly when no abort occurred, the terminal identifier is distinguished, and

$$\mathsf{Win}^{\mathsf{core}}_{\mathsf{PA}} \wedge \neg\mathsf{Complete}(\mathsf{sid}^\star)$$

   holds. Otherwise output 0.

To verify the simulation for a real-distribution input, write $U_i = y_i + sc_i$ and $V_i = e'_i + ec_i$. When $B = as + e$, the residual satisfies

$$\begin{aligned} W_j - aU_i + Bc_i - V_i &= W_j - a(y_i + sc_i) + (as + e)c_i \\ &\quad - (e'_i + ec_i) = W_j - ay_i - e'_i. \end{aligned}$$

Thus the deferred Gaussian pair of the last responder can be identified with (y_i, e'_i). Identifying the deferred value $y_j = y_i$, its response satisfies

$$\begin{aligned} z_j &= y_i + (s - s_K)c_i - m_K \\ &\quad - \sum_{k\in J\setminus\{j\}} z_k + \sum_{k\in J\setminus\{j\}} y_k \\ &= \sum_{k\in J} y_k + (s - s_K)c_i - m_K - \sum_{k\in J\setminus\{j\}} z_k, \end{aligned}$$

which is the prescribed last-responder identity.

The uniform token revealed no information about this deferred pair, and the signer had not previously released a response. Hence this identification remains valid when the final responder is chosen adaptively. Since an identifier has only one activated context, one challenge input closes at most one honest-response vector.

Consequently, a real-distribution input reproduces G_{real}, whereas a uniform-distribution input reproduces G_{uniform}. Let p_{real} and p_{uniform} be their respective probabilities of the terminal decision event. By construction,

$$|p_{\mathsf{real}} - p_{\mathsf{uniform}}| = \mathrm{Adv}^{\mathsf{Hint\text{-}RLWE}}_{\mathcal{B}_{\mathsf{Hint}}} \le \epsilon_{\mathsf{Hint}}.$$

The distinguisher has polynomial overhead under the stated generator and sampler contracts. The earlier label and token simulation errors remain accounted for separately in ϵ_{TD}.

It remains to bound successful outputs in this uniform-key game. Embed a ring-SIS instance (a, B, h) by using its first two coordinates in the verification key and its third as the distinguished label. The simulator for this game can generate

its own hidden dummy (s, e) and all hints, independently of B, and is therefore executable and rewindable without an external hint oracle. Other labels use the stored trapdoors.

For the forking argument, regard the adversary and its simulator as one randomized program $\mathcal{F}$. Separate its non-challenge randomness ρ from the challenge answers $c_1, \ldots, c_Q$. The tape ρ includes the adversary's coins, the label-selection coins, all token and sampler coins, and the locally generated dummy secret (s_0, e_0) and Gaussian hint pairs (y_i, e'_i).

For each distinct challenge input indexed by i, compute

$$U_i = y_i + s_0 c_i, \qquad V_i = e'_i + e_0 c_i.$$

Thus changing a challenge answer also recomputes its associated hint, using the same dummy secret and Gaussian base pair.

On a successful restricted forgery with a previously evaluated terminal challenge input, $\mathcal{F}$ returns its query index and the terminal message–signature pair. Otherwise it returns index zero.

If the terminal challenge input was never evaluated before the output, its answer, sampled during verification, equals the proposed c with probability at most $1/M$. Otherwise, let $k^\star \in [Q]$ be the index of the first evaluation of the terminal challenge input. During the first execution, record a snapshot before answering each new challenge input. The snapshot for $k^\star$ is taken after the input and its associated label have been determined, but before the challenge answer is recorded or returned.

This snapshot contains the adversary and simulator configurations, including any pending oracle invocation and its local variables. It also contains both random-oracle tables, the materialized hint records, all identifier-service tables, token and private-state records, trapdoor records, counters, and random-tape positions. In particular, it preserves all consumption and erasure flags. If the challenge evaluation occurs inside a response query, the snapshot includes that suspended response computation.

Run the second execution with the same non-challenge tape ρ and the same answers $c_1, \ldots, c_{k^\star-1}$. Independently sample new answers $c'_{k^\star}, \ldots, c'_Q$. This can be implemented by restarting $\mathcal{F}$ from its initial state and replaying the common prefix. Equivalently, restore the snapshot and resume the pending challenge evaluation with $c'_{k^\star}$.

Only the prefix tables are restored. All records created after the snapshot in the first execution are discarded before the second continuation. Each continuation maintains its own oracle and service state, and recomputes its hints from its current challenge answers. The no-completion abort rule remains in force in both executions.

Let k' be the terminal challenge index returned by the second execution. Declare the fork successful only if both executions yield successful restricted forgeries, $k' = k^\star$, and $c'_{k^\star} \neq c_{k^\star}$. Otherwise the forking algorithm fails.

Because the executions agree before the $k^\star$th answer, the corresponding query input is identical in both executions. Canonical injective encoding therefore fixes $(\mathsf{vk}, \mathsf{sid}^\dagger, \mathsf{act}^\star, C^\star, \mathsf{msg}^\star)$. The associated label h is already fixed by the internal label-evaluation convention. A successful fork consequently yields two norm-valid signatures for the same complete challenge input, with different challenges. Lemma 4 then extracts a nonzero ring-SIS solution.

All restored state belongs to the reduction's simulated experiment. No rollback interface for the deployed binding service is assumed.

For completeness, let p be the success probability in the uniform-key game restricted to outputs with an already evaluated challenge input. The classical forking inequality gives a solver with success at least

$$p\left(\frac{p}{Q} - \frac{1}{M}\right).$$

Hence $p \le Q/M + \sqrt{Q\epsilon_{\mathsf{SIS}}}$. Adding the unqueried case, undoing the two hybrid changes and the $1/L$ selection, and charging the initial PRF and whole-run sampling errors proves the explicit bound

$$\mathrm{Adv}^{\mathsf{Inc}}_{\Pi_{\mathsf{pp}},\mathcal{A}} \le \epsilon_{\mathsf{PRF}} + \epsilon^{\mathsf{run}}_{\mathsf{samp}} + L\left(\epsilon_{\mathsf{TD}} + \epsilon_{\mathsf{Hint}} + \frac{Q+1}{M} + \sqrt{Q\epsilon_{\mathsf{SIS}}}\right).$$

All losses count the full adversarial interaction; no assumption on a fixed subset or a fixed target message was introduced.□

F. Proof of Lemma 4 (Fixed-Input Extraction with Scaled Lifting)

Proof. In $\mathcal{M}_q$, define

$$X = az + hr - \mathsf{Lift}_1(bc_{\mathbb{Z}}),$$
$$X' = az' + hr' - \mathsf{Lift}_1(bc'_{\mathbb{Z}}),$$
$$\eta_B = B - \mathsf{Lift}_1(b).$$

The lift is an additive homomorphism compatible with multiplication by integer polynomials modulo $q\mathbb{Z}^N$. Thus no equality between elements of different finite rings is assumed. Since $\|\eta_B\|_\infty \le q/(2q_1)$ and both reconstructions equal C,

$$\lfloor X \rceil_{p_2} - \lfloor X' \rceil_{p_2} = \Delta' - \Delta.$$

Applying the rounding error bound to X and X' separately gives

$$\|X - X'\|_2 \le \frac{q}{q_2}\left(2\beta_\Delta + \sqrt{N}\right).$$

All these norms are quotient norms, evaluated using shortest centered representatives; the triangle inequality is valid in the quotient. Let $u = z - z'$, $d = c - c'$, and $v = r - r'$. Then

$$au - Bd + hv = X - X' - \eta_B d \quad \text{in } \mathcal{M}_q.$$

The left side lies in R_q. Since $\|d\|_1 \le 2\tau$, Young's inequality gives

$$\|\eta_B d\|_2 \le \frac{q\tau}{q_1}\sqrt{N}.$$

Take δ to be the centered representative of $-(au - Bd + hv)$. Then

$$\begin{bmatrix} 1 & a & B & h \end{bmatrix} \begin{pmatrix} \delta \\ u \\ -d \\ v \end{pmatrix} = 0, \qquad \|\delta\|_2 \le \beta_\delta.$$

Moreover, $\|u\|_2 \le 2\beta_z$, $\|v\|_2 \le 2\beta_r$, and $\|d\|_2 \le 2\sqrt{\tau}$. The vector is nonzero because $c \ne c'$ as canonically encoded ring elements. Summing these squared bounds gives

$$\begin{aligned}\beta_\delta &= \frac{q}{q_2}\left(2\beta_\Delta + \sqrt{N}\right) + \frac{q\tau}{q_1}\sqrt{N},\\ \|(\delta, u, -d, v)\|_2 &\le \sqrt{4\beta_z^2 + 4\beta_r^2 + 4\tau + \beta_\delta^2}\\ &= \beta_{\mathsf{SIS}}.\end{aligned}$$

□

G. Full Proof of Theorem 3 (State-Assisted Preprocessing-Aware Security)

Proof. The generic lifting theorem bounds the state-assisted PA advantage by the lattice-only incomplete-context advantage. Theorem 2 bounds that advantage by the displayed right-hand side. The target-completion abort in Theorem 2 discards no execution counted by its incomplete-context experiment; the generic binding argument handles completed target identifiers. □

H. Screened Generation and Its Output Law

Put $R = \mathbb{Z}[X]/(X^N+1)$, $d = 2N$, and $\gamma = 1.17\sqrt{q}$. A trial samples independent integer coefficient vectors $f, g \leftarrow D_{\sigma_{\mathsf{td}}}$ before reduction modulo q. Reject unless $f \bmod q$ is invertible. Run a fixed implementation of Falcon's NTRU equation solver on (f, g, q), with a fixed polynomial operation and output-bit-length budget; a timeout or failure rejects the trial.[2] For a returned pair (F, G), verify exactly in R that $fG - gF = q$. Set $h = gf^{-1} \bmod q$, and form the integer coefficient basis $\mathcal{B}$ from the negacyclic rotations of $(g, -f)$ and $(G, -F)$. These vectors generate

$$\Lambda_h = \{(u, v) \in \mathbb{Z}^{2N} : u + hv = 0 \bmod q\}.$$

Indeed their determinant has absolute value q^N, the index of Λ_h, and each vector lies in this kernel. Compute the squared Gram–Schmidt lengths of $\mathcal{B}$ by exact rational arithmetic, or use certified interval upper bounds. Accept only if

$$\max_{i\in[d]} \|\widetilde{b}_i\|_2^2 \le \gamma^2 = \frac{13689}{10000}q.$$

An inconclusive certificate rejects. Thus every accepted basis satisfies the claimed GS bound by construction, without an assumption about the typical norm of sampled (f, g). This is a specification of a certified generator; no implementation or acceptance-rate measurement at TT-128-R1 is claimed here.

Let Acc include every test and resource limit above, and let $p_{\mathsf{acc}} = \Pr[\mathsf{Acc}]$ for an independent trial, including any solver randomness. With a trial cap J_{gen}, return the first accepted $(h, \mathcal{B})$ or $\perp$ if all trials reject. Then

$$\begin{aligned}\Pr[\mathsf{TDGen} \ne \perp] &= 1 - (1 - p_{\mathsf{acc}})^{J_{\mathsf{gen}}},\\ \mathbb{E}[\#\text{trials}] &= \frac{1 - (1 - p_{\mathsf{acc}})^{J_{\mathsf{gen}}}}{p_{\mathsf{acc}}}.\end{aligned}$$

Without the cap the expectation is $1/p_{\mathsf{acc}}$; if one trial has expected cost at most C_{gen}, the expected total cost is at most $C_{\mathsf{gen}}/p_{\mathsf{acc}}$. *Generation assumption.* For the fixed solver, budgets and screening rule, assume a stated inverse-polynomial lower bound $p_{\mathsf{acc}} \ge p_0 > 0$ and polynomial trial cost. No numerical p_0 has been established for TT-128-R1. Bounded-time Gaussian implementations are compared with this ideal sampling experiment using the per-generator error δ_{gen} below. A concrete instantiation must fix the solver version, arithmetic precision, resource limits, trial cap and a justified p_0 before claiming the resulting generation cost. The formulas above do not substitute for these missing concrete choices.

Conditioned on success, the output distribution is exactly the pushforward of the trial distribution conditioned on Acc, independently of the trial cap. In particular, define the public-key law

$$\mathcal{D}_h^{\mathsf{acc}} = \mathcal{L}(gf^{-1} \bmod q \mid \mathsf{Acc}).$$

Screened-key assumption. For every distinguisher of the time and query budget used by the reduction, assume

$$|\Pr[\mathcal{D}(h) = 1 : h \leftarrow \mathcal{D}_h^{\mathsf{acc}}] - \Pr[\mathcal{D}(h) = 1 : h \leftarrow R_q]| \le \epsilon_{\mathsf{key}}.$$

The trapdoor and trial history are not given to this distinguisher. This is an explicit assumption on the *screened* distribution, not a consequence proved here from unscreened NTRU key indistinguishability. Filtering depends on secret coefficients and solver behavior, so even a lower bound on its acceptance probability does not by itself justify transferring pseudorandomness through that filter.

I. Preimages for Every Target and Statistical Bounds

Use the GPV randomized nearest-plane sampler as a reference implementation of Pre.[3] For $w \in R_q$, set $t_w = (\overline{w}, 0) \in \mathbb{Z}^{2N}$. Starting with center t_w, for $i = d, \ldots, 1$ compute the current center's projection coefficient onto $\widetilde{b}_i$, sample an integer with that center and width $\sigma_r/\|\widetilde{b}_i\|_2$, and subtract its multiple of b_i from the current center. If $v \in \Lambda_h$ is the sum of those multiples, output $r = t_w - v$, reduced modulo q only at the protocol interface. Then $f_h(r) = w$ exactly.

For any accepted basis and any w, the target conditional distribution is $D_{t_w+\Lambda_h,\sigma_r}$ on integer vectors. The GPV reference sampler approximates this distribution. Assume a one-dimensional sampler whose distance from its exact shifted discrete Gaussian is at most ν uniformly over every center and width encountered; any finite-precision or truncation errors must be included in this bound. Let $0 < \epsilon < 1$ and require

$$\sigma_r \ge \gamma\sqrt{\frac{\ln(2d(1 + 1/\epsilon))}{\pi}}.$$

The GS smoothing bound gives $\sigma_r \ge \eta_\epsilon(\Lambda_h)$ and every one-dimensional width is at least $\eta_\epsilon(\mathbb{Z})$. Gaussian coset-mass

[2] Falcon specification v1.2, Algorithm 6; https://falcon-sign.info/falcon.pdf. The additional exact certification and resource limits below are part of our specified screening rule, not a claimed benchmark of Falcon at these parameters.

[3] C. Gentry, C. Peikert, and V. Vaikuntanathan, "Trapdoors for hard lattices and new cryptographic constructions," in *Proc. 40th ACM Symp. Theory of Computing (STOC)*, 2008, pp. 197–206; https://doi.org/10.1145/1374376.1374407.

bounds and the sequential sampler's normalization factors then give the conservative bounds

$$\delta_{\mathsf{img}} = \frac{2\epsilon}{1-\epsilon},\ \delta_{\mathsf{GPV}} = \left(\frac{1+\epsilon}{1-\epsilon}\right)^{d} - 1,$$
$$\sup_{w\in R_q} \mathrm{SD}\big(\mathsf{Pre}(h,\mathcal{B},w), D_{t_w+\Lambda_h,\sigma_r}\big) \le \delta_{\mathsf{GPV}} + d\nu.$$

The comparison in the last line is before ring reduction; reduction modulo q cannot increase the distance. For the first bound, every coset mass lies within the usual $(1\pm\epsilon)$ multiples of a common mass, which bounds its normalized image probability relative to uniform. For the second, multiplying the d shifted-Gaussian normalizer ratios bounds the sampler's density ratio by $((1+\epsilon)/(1-\epsilon))^d$; the d approximate scalar calls add at most $d\nu$ by coupling. These estimates are uniform in w, not merely averages over random targets.

For a forward draw $r \leftarrow D^2_{\sigma_r}$, the conditional law given $f_h(r) = w$ is exactly that coset Gaussian. Replacing its image by uniform and then using Pre therefore costs at most

$$\epsilon_{\mathsf{pre}} \le \delta_{\mathsf{img}} + \delta_{\mathsf{GPV}} + d\nu$$

per token, including its later opening. For $d = 8192$ and $\epsilon = 2^{-320}$, a scalar sampler meeting the *design target* $\nu \le 2^{-340}$ would give $\epsilon_{\mathsf{pre}} < 2^{-305}$. This is a conditional numerical budget, not a measurement of an implemented sampler. A faster Falcon sampler can replace the reference algorithm only with a guarantee for the same conditional distribution.

J. Composition over Keys and Tokens

Let $K \le L-1$ count generated nondistinguished key records, and let T_k count replaced tokens under record k, so $\sum_{k=1}^{K} T_k \le T$. Let δ_{gen} bound one implemented capped generator's distance from the ideal capped experiment, including its proposal sampling and numerical errors. Exact GS certification leaves no additional accepted-bad-basis event. Independent attempts give

$$\epsilon_{\mathsf{badTD}} \le K\big((1-p_0)^{J_{\mathsf{gen}}} + \delta_{\mathsf{gen}}\big),$$
$$\epsilon_{\mathsf{TD}} \le K\epsilon_{\mathsf{key}} + \sum_{k=1}^{K} T_k\epsilon_{\mathsf{pre}} + \epsilon_{\mathsf{badTD}}$$
$$\le (L-1)\epsilon_{\mathsf{key}} + T\epsilon_{\mathsf{pre}} + \epsilon_{\mathsf{badTD}}.$$

The term $K\epsilon_{\mathsf{key}}$ accounts for the computational replacement of public labels. The term $\sum_{k=1}^{K} T_k\epsilon_{\mathsf{pre}}$ accounts for the statistical comparison of tokens and their later openings. The term $\epsilon_{\mathsf{badTD}}$ accounts for capped-generation failure and generator implementation error. These contributions are collected in ϵ_{TD}, with generation failure and implementation error counted only once. For $K \ge 1$, choosing $J_{\mathsf{gen}} \ge (\kappa \ln 2 + \ln K)/p_0$ bounds the total ideal generation-failure probability by $2^{-\kappa}$. The per-image uniform bound permits adaptively chosen targets and repeated use of a key. The outer label-guessing factor L in Theorem 2 still applies to ϵ_{TD}. Errors already included in δ_{gen} or $d\nu$ are not charged again in the whole-run error for actual protocol samples.

K. Hardness Assumptions

The theorem is stated using this trapdoor contract, Hint-RLWE, and normalized ring-SIS. It does not require a separate reduction to a particular standard formulation of NTRU if screened-key indistinguishability is adopted explicitly as above. For this proposed NTRU-type generator, however, that is an additional NTRU-like decisional assumption, not something that can be omitted. Search-NTRU key-recovery costs do not alone establish it. Removing this dependency would require another generator satisfying the same contract, or a different simulation proof. The formal scheme's honest algorithms do not themselves invoke the generator, but the present proof needs it to simulate nondistinguished tokens and completed responses.